9
Electrons in a Weak Periodic Potential

One can gain substantial insight into the structure imposed on the electronic energy levels by a periodic potential, if that potential is very weak. This approach might once have been regarded as an instructive, but academic, exercise. We now know, however, that in many cases this apparently unrealistic assumption gives results surprisingly close to the mark. Modern theoretical and experimental studies of the metals found in groups I, II, III, and IV of the periodic table (i.e., metals whose atomic structure consists of *s* and *p* electrons outside of a closed-shell noble gas configuration) indicate that the conduction electrons can be described as moving in what amounts to an almost constant potential. These elements are often referred to as "nearly free electron" metals, because the starting point for their description is the Sommerfeld free electron gas, modified by the presence of a *weak* periodic potential. In this chapter we shall examine some of the broad general features of band structure from the almost free electron point of view. Applications to particular metals will be examined in Chapter 15.

It is by no means obvious why the conduction bands of these metals should be so free-electron-like. There are two fundamental reasons why the strong interactions of the conduction electrons with each other and with the positive ions can have the net effect of a very weak potential.

1. The electron-ion interaction is strongest at small separations, but the conduction electrons are forbidden (by the Pauli principle) from entering the immediate neighborhood of the ions because this region is already occupied by the core electrons.
2. In the region in which the conduction electrons are allowed, their mobility further diminishes the net potential any single electron experiences, for they can *screen* the fields of positively charged ions, diminishing the total effective potential.

These remarks offer only the barest indication of why the following discussion has extensive practical application. We shall return later to the problem of justifying the nearly free electron approach, taking up point 1 in Chapter 11 and point 2 in Chapter 17.

GENERAL APPROACH TO THE SCHRÖDINGER EQUATION WHEN THE POTENTIAL IS WEAK

When the periodic potential is zero, the solutions to Schrödinger's equation are plane waves. A reasonable starting place for the treatment of weak periodic potentials is therefore the expansion of the exact solution in plane waves described in Chapter 8. The wave function of a Bloch level with crystal momentum $\mathbf{k}$ can be written in the form given in Eq. (8.42):

$$\psi_{\mathbf{k}}(\mathbf{r}) = \sum_{\mathbf{K}} c_{\mathbf{k}-\mathbf{K}} e^{i(\mathbf{k}-\mathbf{K})\cdot\mathbf{r}}, \tag{9.1}$$

where the coefficients $c_{\mathbf{k}-\mathbf{K}}$ and the energy of the level ε are determined by the set of Eqs. (8.41):

$$\left[\frac{\hbar^2}{2m}(\mathbf{k}-\mathbf{K})^2 - \varepsilon\right] c_{\mathbf{k}-\mathbf{K}} + \sum_{\mathbf{K}'} U_{\mathbf{K}'-\mathbf{K}} c_{\mathbf{k}-\mathbf{K}'} = 0. \tag{9.2}$$

The sum in (9.1) is over all reciprocal lattice vectors $\mathbf{K}$, and for fixed $\mathbf{k}$ there is an equation of the form (9.2) for each reciprocal lattice vector $\mathbf{K}$. The (infinitely many) different solutions to (9.2) for a given $\mathbf{k}$ are labeled with the band index n. The wave vector $\mathbf{k}$ can (but need not) be considered to lie in the first Brillouin zone of k-space.

In the free electron case, all the Fourier components $U_{\mathbf{K}}$ are precisely zero. Equation (9.2) then becomes

$$(\mathcal{E}^0_{\mathbf{k}-\mathbf{K}} - \mathcal{E})c_{\mathbf{k}-\mathbf{K}} = 0, \tag{9.3}$$

where we have introduced the notation:

$$\mathcal{E}^0_q = \frac{\hbar^2}{2m}q^2. \tag{9.4}$$

Equation (9.3) requires for each $\mathbf{K}$ that either $c_{\mathbf{k}-\mathbf{K}} = 0$ or $\mathcal{E} = \mathcal{E}^0_{\mathbf{k}-\mathbf{K}}$. The latter possibility can occur for only a single $\mathbf{K}$, unless it happens that some of the $\mathcal{E}^0_{\mathbf{k}-\mathbf{K}}$ are equal for several different choices of $\mathbf{K}$. If such degeneracy does *not* occur, then we have the expected class of free electron solutions:

$$\mathcal{E} = \mathcal{E}^0_{\mathbf{k}-\mathbf{K}}, \quad \psi_{\mathbf{k}} \propto e^{i(\mathbf{k}-\mathbf{K})\cdot\mathbf{r}}. \tag{9.5}$$

If, however, there is a group of reciprocal lattice vectors $\mathbf{K}_1, \ldots, \mathbf{K}_m$ satisfying

$$\mathcal{E}^0_{\mathbf{k}-\mathbf{K}_1} = \cdots = \mathcal{E}^0_{\mathbf{k}-\mathbf{K}_m}, \tag{9.6}$$

then when $\mathcal{E}$ is equal to the common value of these free electron energies there are m independent degenerate plane wave solutions. Since any linear combination of degenerate solutions is also a solution, one has complete freedom in choosing the coefficients $c_{\mathbf{k}-\mathbf{K}}$ for $\mathbf{K} = \mathbf{K}_1, \ldots, \mathbf{K}_m$.

These simple observations acquire more substance when the $U_{\mathbf{K}}$ are not zero, but very small. The analysis still divides naturally into two cases, corresponding to the nondegenerate and degenerate cases for free electrons. Now, however, the basis for the distinction is not the exact equality[1] of two or more distinct free electron levels, but only whether they are equal aside from terms of order U.

Case 1 Fix $\mathbf{k}$ and consider a particular reciprocal lattice vector $\mathbf{K}_1$ such that the free electron energy $\mathcal{E}^0_{\mathbf{k}-\mathbf{K}_1}$ is far from the values of $\mathcal{E}^0_{\mathbf{k}-\mathbf{K}}$ (for all other $\mathbf{K}$) compared with U (see Figure 9.1)[2]:

$$|\mathcal{E}^0_{\mathbf{k}-\mathbf{K}_1} - \mathcal{E}^0_{\mathbf{k}-\mathbf{K}}| \gg U, \qquad \text{for fixed } \mathbf{k} \text{ and all } \mathbf{K} \neq \mathbf{K}_1. \tag{9.7}$$

We wish to investigate the effect of the potential on that free electron level given by:

$$\mathcal{E} = \mathcal{E}^0_{\mathbf{k}-\mathbf{K}_1}, \quad c_{\mathbf{k}-\mathbf{K}} = 0, \qquad \mathbf{K} \neq \mathbf{K}_1. \tag{9.8}$$

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[1] The reader familiar with stationary perturbation theory may think that if there is no *exact* degeneracy, we can always make all level differences large compared with U by considering sufficiently small U. That is indeed true *for any given* $\mathbf{k}$. However, once we are given a definite U, no matter how small, we want a procedure valid for all $\mathbf{k}$ in the first Brillouin zone. We shall see that no matter how small U is we can always find some values of $\mathbf{k}$ for which the unperturbed levels are closer together than U. Therefore what we are doing is more subtle than conventional degenerate perturbation theory.

[2] In inequalities of this form we shall use U to refer to a typical Fourier component of the potential.

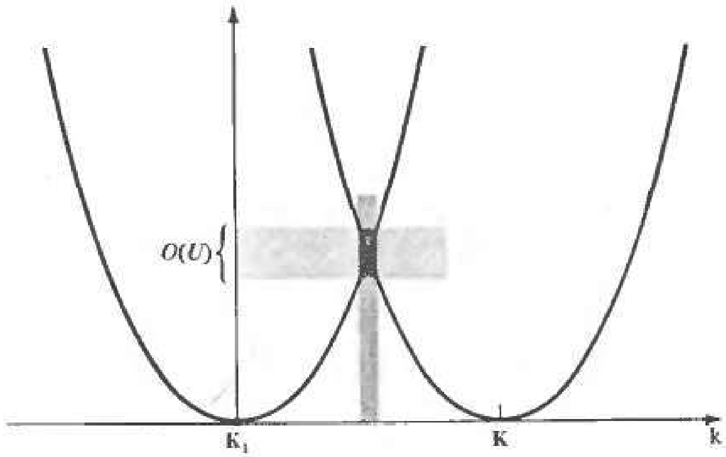

Figure 9.1
For the range of $\mathbf{k}$ within limits indicated by the dark band the free electron levels $\mathcal{E}_{\mathbf{k}-\mathbf{K}_1}$ and $\mathcal{E}_{\mathbf{k}-\mathbf{K}}$ differ by an energy $O(U)$.

Setting $\mathbf{K} = \mathbf{K}_1$ in Eq. (9.2) (and using the short notation (9.4)) we have (dropping the prime from the summation index):

$$(\mathcal{E} - \mathcal{E}^0_{\mathbf{k}-\mathbf{K}_1})c_{\mathbf{k}-\mathbf{K}_1} = \sum_{\mathbf{K}} U_{\mathbf{K}-\mathbf{K}_1}c_{\mathbf{k}-\mathbf{K}}. \tag{9.9}$$

Because we have picked the additive constant in the potential energy so that $U_{\mathbf{K}} = 0$ when $\mathbf{K} = \mathbf{0}$ (see page 137), only terms with $\mathbf{K} \neq \mathbf{K}_1$ appear on the right-hand side of (9.9). Since we are examining that solution for which $c_{\mathbf{k}-\mathbf{K}}$ vanishes when $\mathbf{K} \neq \mathbf{K}_1$ in the limit of vanishing U, we expect the right-hand side of (9.9) to be of second order in U. This can be explicitly confirmed by writing Eq. (9.2) for $\mathbf{K} \neq \mathbf{K}_1$ as

$$c_{\mathbf{k}-\mathbf{K}} = \frac{U_{\mathbf{K}_1-\mathbf{K}}c_{\mathbf{k}-\mathbf{K}_1}}{\mathcal{E} - \mathcal{E}^0_{\mathbf{k}-\mathbf{K}}} + \sum_{\mathbf{K}'\neq\mathbf{K}_1} \frac{U_{\mathbf{K}'-\mathbf{K}}c_{\mathbf{k}-\mathbf{K}'}}{\mathcal{E} - \mathcal{E}^0_{\mathbf{k}-\mathbf{K}}}. \tag{9.10}$$

We have separated out of the sum in (9.10) the term containing $c_{\mathbf{k}-\mathbf{K}_1}$, since it will be an order of magnitude larger than the remaining terms, which involve $c_{\mathbf{k}-\mathbf{K}'}$ for $\mathbf{K}' \neq \mathbf{K}_1$. This conclusion depends on the assumption (9.7) that the level $\mathcal{E}^0_{\mathbf{k}-\mathbf{K}_1}$ is not nearly degenerate to some other $\mathcal{E}^0_{\mathbf{k}-\mathbf{K}}$. Such a near degeneracy could cause some of the denominators in (9.10) to be of order U, canceling the explicit U in the numerator and resulting in additional terms in the sum in (9.10) comparable to the $\mathbf{K} = \mathbf{K}_1$ term.

Therefore, provided there is no near degeneracy,

$$c_{\mathbf{k}-\mathbf{K}} = \frac{U_{\mathbf{K}_1-\mathbf{K}}c_{\mathbf{k}-\mathbf{K}_1}}{\mathcal{E} - \mathcal{E}^0_{\mathbf{k}-\mathbf{K}}} + O(U^2). \tag{9.11}$$

Placing this in (9.9), we find:

$$(\mathcal{E} - \mathcal{E}^0_{\mathbf{k}-\mathbf{K}_1})c_{\mathbf{k}-\mathbf{K}_1} = \sum_{\mathbf{K}} \frac{U_{\mathbf{K}-\mathbf{K}_1}U_{\mathbf{K}_1-\mathbf{K}}}{\mathcal{E} - \mathcal{E}^0_{\mathbf{k}-\mathbf{K}}}c_{\mathbf{k}-\mathbf{K}_1} + O(U^3). \tag{9.12}$$

Thus the perturbed energy level $\mathcal{E}$ differs from the free electron value $\mathcal{E}^0_{\mathbf{k}-\mathbf{K}_1}$ by terms of order U^2. To solve Eq. (9.12) for $\mathcal{E}$ to this order, it therefore suffices to replace the $\mathcal{E}$ appearing in the denominator on the right-hand side by $\mathcal{E}^0_{\mathbf{k}-\mathbf{K}_1}$, leading to the following expression[3] for $\mathcal{E}$, correct to second order in U:

$$\mathcal{E} = \mathcal{E}^0_{\mathbf{k}-\mathbf{K}_1} + \sum_{\mathbf{K}} \frac{|U_{\mathbf{K}-\mathbf{K}_1}|^2}{\mathcal{E}^0_{\mathbf{k}-\mathbf{K}_1} - \mathcal{E}^0_{\mathbf{k}-\mathbf{K}}} + O(U^3). \tag{9.13}$$

Equation (9.13) asserts that weakly perturbed nondegenerate bands repel each other, for every level $\mathcal{E}^0_{\mathbf{k}-\mathbf{K}}$ that lies below $\mathcal{E}^0_{\mathbf{k}-\mathbf{K}_1}$ contributes a term in (9.13) that raises the value of $\mathcal{E}$, while every level that lies above $\mathcal{E}^0_{\mathbf{k}-\mathbf{K}_1}$ contributes a term that lowers the energy. However, the most important feature to emerge from this analysis of the case of no near degeneracy, is simply the gross observation that the shift in energy from the free electron value is second order in U. In the nearly degenerate case (as we shall now see) the shift in energy can be linear in U. Therefore, to leading order in the weak periodic potential, it is only the nearly degenerate free electron levels that are significantly shifted, and we must devote most of our attention to this important case.

Case 2 Suppose the value of $\mathbf{k}$ is such that there are reciprocal lattice vectors $\mathbf{K}_1, \ldots, \mathbf{K}_m$ with $\mathcal{E}^0_{\mathbf{k}-\mathbf{K}_1}, \ldots, \mathcal{E}^0_{\mathbf{k}-\mathbf{K}_m}$ all within order U of each other,[4] but far apart from the other $\mathcal{E}^0_{\mathbf{k}-\mathbf{K}}$ on the scale of U:

$$|\mathcal{E}^0_{\mathbf{k}-\mathbf{K}} - \mathcal{E}^0_{\mathbf{k}-\mathbf{K}_i}| \gg U, \qquad i = 1, \ldots, m, \qquad \mathbf{K} \neq \mathbf{K}_1, \ldots, \mathbf{K}_m. \tag{9.14}$$

In this case we must treat separately those equations given by (9.2) when $\mathbf{K}$ is set equal to any of the m values $\mathbf{K}_1, \ldots, \mathbf{K}_m$. This gives m equations corresponding to the single equation (9.9) in the nondegenerate case. In these m equations we separate from the sum those terms containing the coefficients $c_{\mathbf{k}-\mathbf{K}_j}$, $j = 1, \ldots, m$, which need not be small in the limit of vanishing interaction, from the remaining $c_{\mathbf{k}-\mathbf{K}}$, which will be at most of order U. Thus we have

$$(\mathcal{E} - \mathcal{E}^0_{\mathbf{k}-\mathbf{K}_i})c_{\mathbf{k}-\mathbf{K}_i} = \sum_{j=1}^{m} U_{\mathbf{K}_j-\mathbf{K}_i} c_{\mathbf{k}-\mathbf{K}_j} + \sum_{\mathbf{K}\neq\mathbf{K}_1,\ldots,\mathbf{K}_m} U_{\mathbf{K}-\mathbf{K}_i} c_{\mathbf{k}-\mathbf{K}}, \qquad i = 1, \ldots, m. \tag{9.15}$$

Making the same separation in the sum, we can write Eq. (9.2) for the remaining levels as

$$c_{\mathbf{k}-\mathbf{K}} = \frac{1}{\mathcal{E} - \mathcal{E}^0_{\mathbf{k}-\mathbf{K}}}\left(\sum_{j=1}^{m} U_{\mathbf{K}_j-\mathbf{K}} c_{\mathbf{k}-\mathbf{K}_j} + \sum_{\mathbf{K}'\neq\mathbf{K}_1,\ldots,\mathbf{K}_m} U_{\mathbf{K}'-\mathbf{K}} c_{\mathbf{k}-\mathbf{K}'}\right), \qquad \mathbf{K} \neq \mathbf{K}_1, \ldots, \mathbf{K}_m, \tag{9.16}$$

(which corresponds to equation (9.10) in the case of no near degeneracy).

Since $c_{\mathbf{k}-\mathbf{K}}$ will be at most of order U when $\mathbf{K} \neq \mathbf{K}_1, \ldots, \mathbf{K}_m$, Eq. (9.16) gives

$$c_{\mathbf{k}-\mathbf{K}} = \frac{1}{\mathcal{E} - \mathcal{E}^0_{\mathbf{k}-\mathbf{K}}} \sum_{j=1}^{m} U_{\mathbf{K}_j-\mathbf{K}} c_{\mathbf{k}-\mathbf{K}_j} + O(U^2). \tag{9.17}$$

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[3] We use Eq. (8.34), $U_{-\mathbf{K}} = U_{\mathbf{K}}^*$.

[4] In one dimension m cannot exceed 2, but in three dimensions it can be quite large.

Placing this in (9.15), we find that

$$(\mathcal{E} - \mathcal{E}^0_{\mathbf{k}-\mathbf{K}_i})c_{\mathbf{k}-\mathbf{K}_i} = \sum_{j=1}^{m} U_{\mathbf{K}_j-\mathbf{K}_i} c_{\mathbf{k}-\mathbf{K}_j} + \sum_{j=1}^{m} \left(\sum_{\mathbf{K} \neq \mathbf{K}_1, \ldots, \mathbf{K}_m} \frac{U_{\mathbf{K}-\mathbf{K}_i} U_{\mathbf{K}_j-\mathbf{K}}}{\mathcal{E} - \mathcal{E}^0_{\mathbf{k}-\mathbf{K}}} \right) c_{\mathbf{k}-\mathbf{K}_j} + O(U^3). \quad (9.18)$$

Compare this with the result (9.12) in the case of no near degeneracy. There we found an explicit expression for the shift in energy to order U^2 (to which the set of equations (9.18) reduces when $m = 1$). Now, however, we find that to an accuracy of order U^2 the determination of the shifts in the m nearly degenerate levels reduces to the solution of m coupled equations[5] for the $c_{\mathbf{k}-\mathbf{K}_i}$. Furthermore, the coefficients in the second term on the right-hand side of these equations are of higher order in U than those in the first.[6] Consequently, to find the *leading* corrections in U we can replace (9.18) by the far simpler equations:

$$(\mathcal{E} - \mathcal{E}^0_{\mathbf{k}-\mathbf{K}_i})c_{\mathbf{k}-\mathbf{K}_i} = \sum_{j=1}^{m} U_{\mathbf{K}_j-\mathbf{K}_i} c_{\mathbf{k}-\mathbf{K}_j}, \qquad i = 1, \ldots, m, \quad (9.19)$$

which are just the general equations for a system of m quantum levels.[7]

ENERGY LEVELS NEAR A SINGLE BRAGG PLANE

The simplest and most important example of the preceding discussion is when two free electron levels are within order U of each other, but far compared with U from all other levels. When this happens, Eq. (9.19) reduces to the two equations:

$$\begin{aligned}(\mathcal{E} - \mathcal{E}^0_{\mathbf{k}-\mathbf{K}_1})c_{\mathbf{k}-\mathbf{K}_1} &= U_{\mathbf{K}_2-\mathbf{K}_1} c_{\mathbf{k}-\mathbf{K}_2},\\ (\mathcal{E} - \mathcal{E}^0_{\mathbf{k}-\mathbf{K}_2})c_{\mathbf{k}-\mathbf{K}_2} &= U_{\mathbf{K}_1-\mathbf{K}_2} c_{\mathbf{k}-\mathbf{K}_1}.\end{aligned} \quad (9.20)$$

When only two levels are involved, there is little point in continuing with the notational convention that labels them symmetrically. We therefore introduce variables particularly convenient for the two-level problem:

$$\mathbf{q} = \mathbf{k} - \mathbf{K}_1 \quad \text{and} \quad \mathbf{K} = \mathbf{K}_2 - \mathbf{K}_1, \quad (9.21)$$

and write (9.20) as

$$\begin{aligned}(\mathcal{E} - \mathcal{E}^0_{\mathbf{q}})c_{\mathbf{q}} &= U_{\mathbf{K}} c_{\mathbf{q}-\mathbf{K}},\\ (\mathcal{E} - \mathcal{E}^0_{\mathbf{q}-\mathbf{K}})c_{\mathbf{q}-\mathbf{K}} &= U_{-\mathbf{K}} c_{\mathbf{q}} = U^*_{\mathbf{K}} c_{\mathbf{q}}.\end{aligned} \quad (9.22)$$

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[5] These are rather closely related to the equations of *second-order degenerate* perturbation theory, to which they reduce when all the $\mathcal{E}^0_{\mathbf{k}-\mathbf{K}_i}$ are rigorously equal, $i = 1, \ldots, m$. (See L. D. Landau and E. M. Lifshitz, *Quantum Mechanics*, Addison-Wesley, Reading Mass., 1965, p. 134.)

[6] The numerator is explicitly of order U^2, and since only $\mathbf{K}$-values different from $\mathbf{K}_1, \ldots, \mathbf{K}_m$ appear in the sum, the denominator is not of order U when $\mathcal{E}$ is close to the $\mathcal{E}^0_{\mathbf{k}-\mathbf{K}_i}$, $i = 1, \ldots, m$.

[7] Note that the rule of thumb for going from (9.19) back to the more accurate form in (9.18) is simply that U should be replaced by U', where

$$U'_{\mathbf{K}_j-\mathbf{K}_i} = U_{\mathbf{K}_j-\mathbf{K}_i} + \sum_{\mathbf{K} \neq \mathbf{K}_1, \ldots, \mathbf{K}_m} \frac{U_{\mathbf{K}_j-\mathbf{K}} U_{\mathbf{K}-\mathbf{K}_i}}{\mathcal{E} - \mathcal{E}^0_{\mathbf{k}-\mathbf{K}}}.$$

We have:

$$\mathcal{E}^0_{\mathbf{q}} \approx \mathcal{E}^0_{\mathbf{q}-\mathbf{K}}, \quad \left|\mathcal{E}^0_{\mathbf{q}} - \mathcal{E}^0_{\mathbf{q}-\mathbf{K}'}\right| \gg U, \qquad \text{for } \mathbf{K}' \neq \mathbf{K}, \mathbf{0}. \tag{9.23}$$

Now $\mathcal{E}^0_{\mathbf{q}}$ is equal to $\mathcal{E}^0_{\mathbf{q}-\mathbf{K}}$ for some reciprocal lattice vector only when $|\mathbf{q}| = |\mathbf{q} - \mathbf{K}|$. This means (Figure 9.2a) that $\mathbf{q}$ must lie on the Bragg plane (see Chapter 6) bisecting the line joining the origin of $\mathbf{k}$ space to the reciprocal lattice point $\mathbf{K}$. The assertion that $\mathcal{E}^0_{\mathbf{q}} = \mathcal{E}_{\mathbf{q}-\mathbf{K}'}$ only for $\mathbf{K}' = \mathbf{K}$ requires that $\mathbf{q}$ lie *only* on this Bragg plane, and on no other.

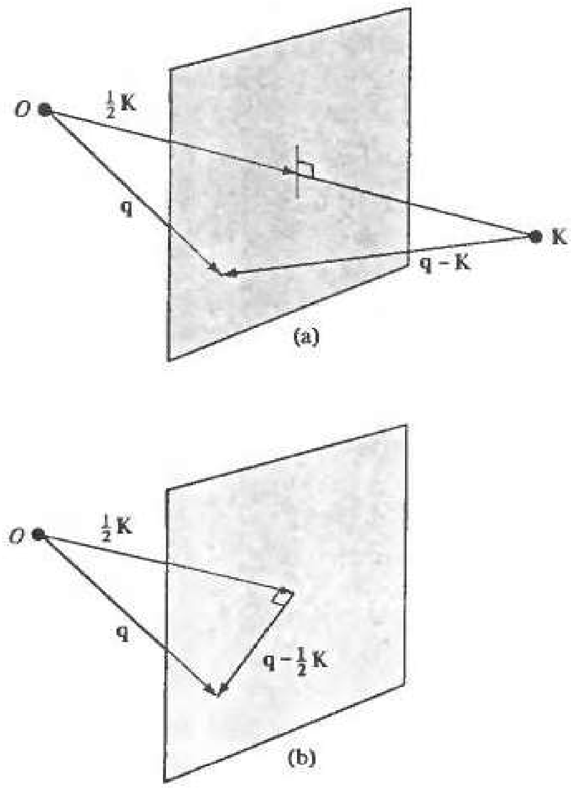

Figure 9.2
(a) If $|\mathbf{q}| = |\mathbf{q} - \mathbf{K}|$, then the point $\mathbf{q}$ must lie in the Bragg plane determined by $\mathbf{K}$. (b) If the point $\mathbf{q}$ lies in the Bragg plane, then the vector $\mathbf{q} - \frac{1}{2}\mathbf{K}$ is parallel to the plane.

Thus conditions (9.23) have the geometric significance of requiring $\mathbf{q}$ to be close to a Bragg plane (but not close to a place where *two* or more Bragg planes intersect). Therefore the case of two nearly degenerate levels applies to an electron whose wave vector very nearly satisfies the condition for a single Bragg scattering.[8] Correspondingly, the general case of many nearly degenerate levels applies to the treatment of a free electron level whose wave vector is close to one at which many simultaneous Bragg reflections can occur. Since the nearly degenerate levels are the most strongly affected by a weak periodic potential, we conclude that *a weak periodic potential has its major effects on only those free electron levels whose wave vectors are close to ones at which Bragg reflections can occur.*

We discuss systematically on pages 162 to 166 when free electron wave vectors do, or do not, lie on Bragg planes, as well as the general structure this imposes on the energy levels in a weak potential. First, however, we examine the level structure

[8] An incident X-ray beam undergoes Bragg reflection only if its wave vector lies on a Bragg plane (see Chapter 6).

when only a single Bragg plane is nearby, as determined by (9.22). These equations have a solution when:

$$\begin{vmatrix} \mathcal{E} - \mathcal{E}^0_{\mathbf{q}} & -U_{\mathbf{K}} \\ -U^*_{\mathbf{K}} & \mathcal{E} - \mathcal{E}^0_{\mathbf{q}-\mathbf{K}} \end{vmatrix} = 0. \tag{9.24}$$

This leads to a quadratic equation

$$(\mathcal{E} - \mathcal{E}^0_{\mathbf{q}})(\mathcal{E} - \mathcal{E}^0_{\mathbf{q}-\mathbf{K}}) = |U_{\mathbf{K}}|^2. \tag{9.25}$$

The two roots

$$\mathcal{E} = \tfrac{1}{2}(\mathcal{E}^0_{\mathbf{q}} + \mathcal{E}^0_{\mathbf{q}-\mathbf{K}}) \pm \left[\left(\frac{\mathcal{E}^0_{\mathbf{q}} - \mathcal{E}^0_{\mathbf{q}-\mathbf{K}}}{2}\right)^2 + |U_{\mathbf{K}}|^2\right]^{1/2} \tag{9.26}$$

give the dominant effect of the periodic potential on the energies of the two free electron levels $\mathcal{E}^0_{\mathbf{q}}$ and $\mathcal{E}^0_{\mathbf{q}-\mathbf{K}}$ when $\mathbf{q}$ is close to the Bragg plane determined by $\mathbf{K}$. These are plotted in Figure 9.3.

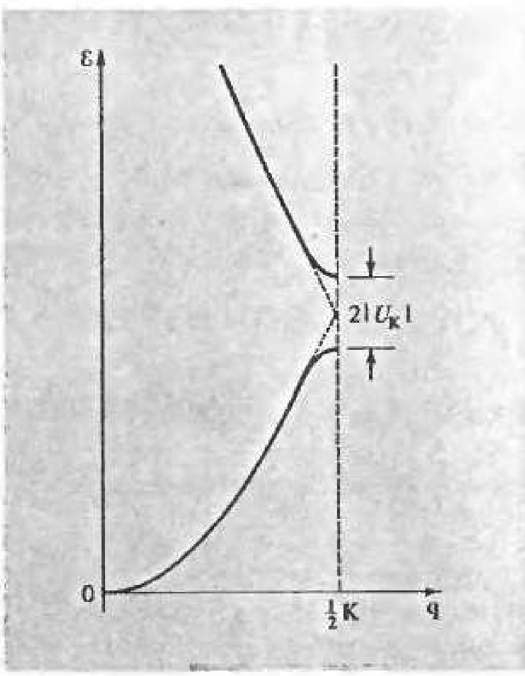

Figure 9.3
Plot of the energy bands given by Eq. (9.26) for $\mathbf{q}$ parallel to $\mathbf{K}$. The lower band corresponds to the choice of a minus sign in (9.26) and the upper band to a plus sign. When $\mathbf{q} = \frac{1}{2}\mathbf{K}$, the two bands are separated by a band gap of magnitude $2|U_{\mathbf{K}}|$. When $\mathbf{q}$ is far removed from the Bragg plane, the levels (to leading order) are indistinguishable from their free electron values (denoted by dotted lines).

The result (9.26) is particularly simple for points lying *on* the Bragg plane since, when $\mathbf{q}$ is on the Bragg plane, $\mathcal{E}^0_{\mathbf{q}} = \mathcal{E}^0_{\mathbf{q}-\mathbf{K}}$. Hence

$$\mathcal{E} = \mathcal{E}^0_{\mathbf{q}} \pm |U_{\mathbf{K}}|, \quad \mathbf{q} \text{ on a single Bragg plane.} \tag{9.27}$$

Thus, at all points on the Bragg plane, one level is uniformly raised by $|U_{\mathbf{K}}|$ and the other is uniformly lowered by the same amount.

It is also easily verified from (9.26) that when $\mathcal{E}^0_{\mathbf{q}} = \mathcal{E}^0_{\mathbf{q}-\mathbf{K}}$,

$$\frac{\partial \mathcal{E}}{\partial \mathbf{q}} = \frac{\hbar^2}{m}(\mathbf{q} - \tfrac{1}{2}\mathbf{K}); \tag{9.28}$$

i.e., when the point $\mathbf{q}$ is on the Bragg plane the gradient of $\mathcal{E}$ is parallel to the plane (see Figure 9.2b). Since the gradient is perpendicular to the surfaces on which a function is constant, the constant-energy surfaces at the Bragg plane are perpendicular to the plane.[9]

When $\mathbf{q}$ lies on a single Bragg plane we may also easily determine the form of the wave functions corresponding to the two solutions $\mathcal{E} = \mathcal{E}^0_{\mathbf{q}} \pm |U_{\mathbf{K}}|$. From (9.22), when $\mathcal{E}$ is given by (9.27), the two coefficients $c_{\mathbf{q}}$ and $c_{\mathbf{q}-\mathbf{K}}$ satisfy[10]

$$c_{\mathbf{q}} = \pm \operatorname{sgn}(U_{\mathbf{K}}) c_{\mathbf{q}-\mathbf{K}}. \tag{9.29}$$

Since these two coefficients are the dominant ones in the plane-wave expansion (9.1), it follows that if $U_{\mathbf{K}} > 0$, then

$$\begin{aligned} |\psi(\mathbf{r})|^2 &\propto (\cos \tfrac{1}{2}\mathbf{K}\cdot\mathbf{r})^2, \quad \mathcal{E} = \mathcal{E}^0_{\mathbf{q}} + |U_{\mathbf{K}}|, \\ |\psi(\mathbf{r})|^2 &\propto (\sin \tfrac{1}{2}\mathbf{K}\cdot\mathbf{r})^2, \quad \mathcal{E} = \mathcal{E}^0_{\mathbf{q}} - |U_{\mathbf{K}}|, \end{aligned}$$

while if $U_{\mathbf{K}} < 0$, then

$$\begin{aligned} |\psi(\mathbf{r})|^2 &\propto (\sin \tfrac{1}{2}\mathbf{K}\cdot\mathbf{r})^2, \quad \mathcal{E} = \mathcal{E}^0_{\mathbf{q}} + |U_{\mathbf{K}}|, \\ |\psi(\mathbf{r})|^2 &\propto (\cos \tfrac{1}{2}\mathbf{K}\cdot\mathbf{r})^2, \quad \mathcal{E} = \mathcal{E}^0_{\mathbf{q}} - |U_{\mathbf{K}}|. \end{aligned} \tag{9.30}$$

Sometimes the two types of linear combination are called "p-like" ($|\psi|^2 \sim \sin^2 \frac{1}{2}\mathbf{K}\cdot\mathbf{r}$) and "$s$-like" ($|\psi|^2 \sim \cos^2 \frac{1}{2}\mathbf{K}\cdot\mathbf{r}$) because of their position dependence near lattice points. The s-like combination, like an atomic s-level, does not vanish at the ion; in the p-like combination the charge density vanishes as the square of the distance from the ion for small distances, which is also a characteristic of atomic p-levels.

ENERGY BANDS IN ONE DIMENSION

We can illustrate these general conclusions in one dimension, where twofold degeneracy is the most that can ever occur. In the absence of any interaction the electronic energy levels are just a parabola in k (Figure 9.4a). To leading order in the weak one-dimensional periodic potential this curve remains correct except near Bragg "planes" (which are points in one dimension). When q is near a Bragg "plane" corresponding to the reciprocal lattice vector K (i.e., the point $\frac{1}{2}K$) the corrected energy levels are determined by drawing another free electron parabola centered around K (Figure 9.4b), noting that the degeneracy at the point of intersection is split by $2|U_K|$ in such a way that both curves have zero slope at that point, and redrawing Figure 9.4b to get Figure 9.4c. The original free electron curve is therefore modified as in Figure 9.4d. When all Bragg planes and their associated Fourier components are included, we end up with a set of curves such as those shown in Figure 9.4e. This particular way of depicting the energy levels is known as the *extended-zone scheme*.

If we insist on specifying all the levels by a wave vector k in the first Brillouin zone, then we must translate the pieces of Figure 9.4e, through reciprocal lattice vectors, into the first Brillouin zone. The result is shown in Figure 9.4f. The representation is that of the *reduced-zone scheme* (see page 142).

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[9] This result is often, but not always, true even when the periodic potential is not weak, because the Bragg planes occupy positions of fairly high symmetry.

[10] For simplicity we assume here that $U_{\mathbf{K}}$ is real (the crystal has inversion symmetry).

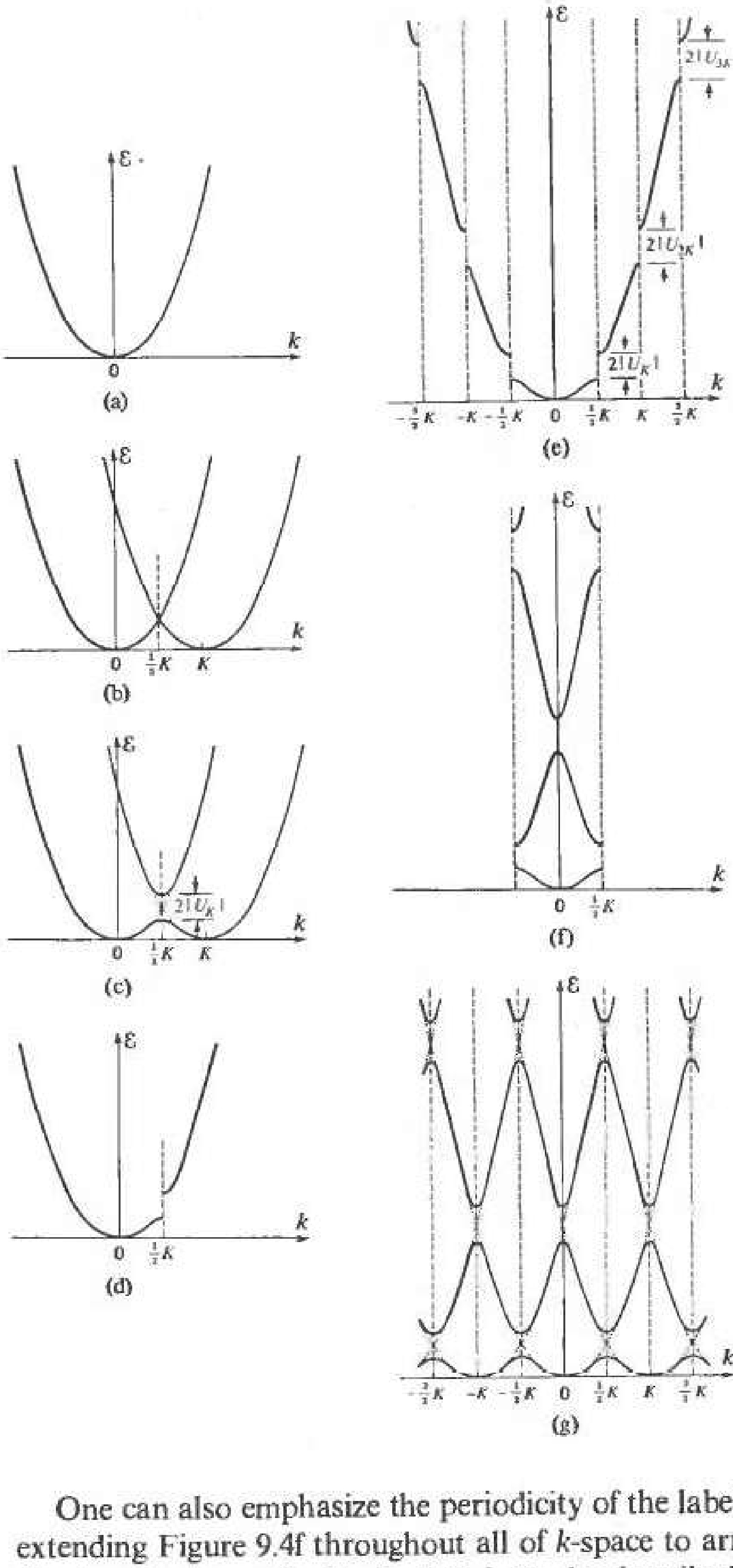

Figure 9.4
(a) The free electron $\mathcal{E}$ vs. k parabola in one dimension. (b) Step 1 in the construction to determine the distortion in the free electron parabola in the neighborhood of a Bragg "plane," due to a weak periodic potential. If the Bragg "plane" is that determined by K, a second free electron parabola is drawn, centered on K. (c) Step 2 in the construction to determine the distortion in the free electron parabola in the neighborhood of a Bragg "plane." The degeneracy of the two parabolas at $K/2$ is split. (d) Those portions of part (c) corresponding to the original free electron parabola given in (a). (e) Effect of all additional Bragg "planes" on the free electron parabola. This particular way of displaying the electronic levels in a periodic potential is known as the *extended-zone scheme*. (f) The levels of (e), displayed in a *reduced-zone scheme*. (g) Free electron levels of (e) or (f) in a *repeated-zone scheme*.

One can also emphasize the periodicity of the labeling in k-space by periodically extending Figure 9.4f throughout all of k-space to arrive at Figure 9.4g, which emphasizes that a particular level at k can be described by any wave vector differing from k by a reciprocal lattice vector. This representation is the *repeated-zone scheme* (see page 142). The reduced-zone scheme indexes each level with a k lying in the first zone, while the extended-zone scheme uses a labeling emphasizing continuity with the free electron levels. The repeated-zone scheme is the most general representation, but is highly redundant, since the same level is shown many times, for all equivalent wave vectors $k, k \pm K, k \pm 2K, \ldots$.

ENERGY-WAVE-VECTOR CURVES IN THREE DIMENSIONS

In three dimensions the structure of the energy bands is sometimes displayed by plotting $\mathcal{E}$ vs. $\mathbf{k}$ along particular straight lines in k-space. Such curves are generally shown in a reduced-zone scheme, since for general directions in k-space they are not periodic. Even in the completely free electron approximation these curves are surprisingly complex. An example is shown in Figure 9.5, which was constructed by plotting, as $\mathbf{k}$ varied along the particular lines shown, the values of $\mathcal{E}^0_{\mathbf{k}-\mathbf{K}} = \hbar^2(\mathbf{k} - \mathbf{K})^2/2m$ for all reciprocal lattice vectors $\mathbf{K}$ close enough to the origin to lead to energies lower than the top of the vertical scale.

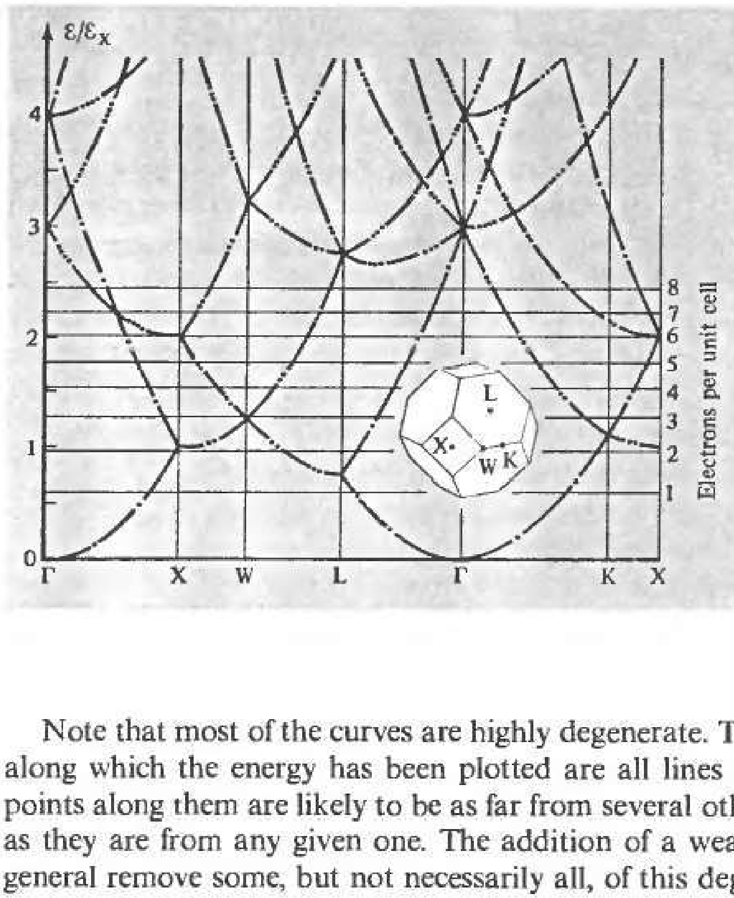

Figure 9.5
Free electron energy levels for an fcc Bravais lattice. The energies are plotted along lines in the first Brillouin zone joining the points $\Gamma(\mathbf{k} = 0)$, K, L, W, and X. $\mathcal{E}_x$ is the energy at point X ($[\hbar^2/2m][2\pi/a]^2$). The horizontal lines give Fermi energies for the indicated numbers of electrons per primitive cell. The number of dots on a curve specifies the number of degenerate free electron levels represented by the curve. (From F. Herman, in *An Atomistic Approach to the Nature and Properties of Materials*, J. A. Pask, ed., Wiley, New York, 1967.)

Note that most of the curves are highly degenerate. This is because the directions along which the energy has been plotted are all lines of fairly high symmetry, so points along them are likely to be as far from several other reciprocal lattice vectors as they are from any given one. The addition of a weak periodic potential will in general remove some, but not necessarily all, of this degeneracy. The mathematical theory of groups is often used to determine how such degeneracies will be split.

THE ENERGY GAP

Quite generally, a weak periodic potential introduces an "energy gap" at Bragg planes. By this we mean the following:

When $U_{\mathbf{K}} = 0$, as $\mathbf{k}$ crosses a Bragg plane the energy changes continuously from the lower root of (9.26) to the upper, as illustrated in Figure 9.4b. When $U_{\mathbf{K}} \neq 0$, this is no longer so. The energy only changes continuously with $\mathbf{k}$, as the Bragg plane is crossed, if one stays with the lower (or upper) root, as illustrated in Figure 9.4c. To change branches as $\mathbf{k}$ varies continuously it is now necessary for the energy to change *discontinuously* by at least $2|U_{\mathbf{K}}|$.

We shall see in Chapter 12 that this mathematical separation of the two bands is reflected in a physical separation: When the action of an external field changes an electron's wave vector, the presence of the energy gap requires that upon crossing the Bragg plane, the electron must emerge in a level whose energy remains in the original branch of $\mathcal{E}(\mathbf{k})$. It is this property that makes the energy gap of fundamental importance in electronic transport properties.

BRILLOUIN ZONES

Using the theory of electrons in a weak periodic potential to determine the complete band structure of a three-dimensional crystal leads to geometrical constructions of great complexity. It is often most important to determine the Fermi surface (page 141) and the behavior of the $\mathcal{E}_n(\mathbf{k})$ in its immediate vicinity.

In doing this for weak potentials, the procedure is first to draw the free electron Fermi sphere centered at $\mathbf{k} = 0$. Next, one notes that the sphere will be deformed in a manner of which Figure 9.6 is characteristic[11] when it crosses a Bragg plane and in a correspondingly more complex way when it passes near several Bragg planes. When the effects of all Bragg planes are inserted, this leads to a representation of the Fermi surface as a fractured sphere in the extended-zone scheme. To construct the portions of the Fermi surface lying in the various bands in the repeated-zone scheme one can make a similar construction, starting with free electron spheres centered about all reciprocal lattice points. To construct the Fermi surface in the reduced-zone scheme, one can translate all the pieces of the single fractured sphere back into the first zone through reciprocal lattice vectors. This procedure is made systematic through the geometrical notion of the higher Brillouin zones.

Recall that the first Brillouin zone is the Wigner-Seitz primitive cell of the reciprocal lattice (pages 73 and 89), i.e. the set of points lying closer to $\mathbf{K} = 0$ than to any other

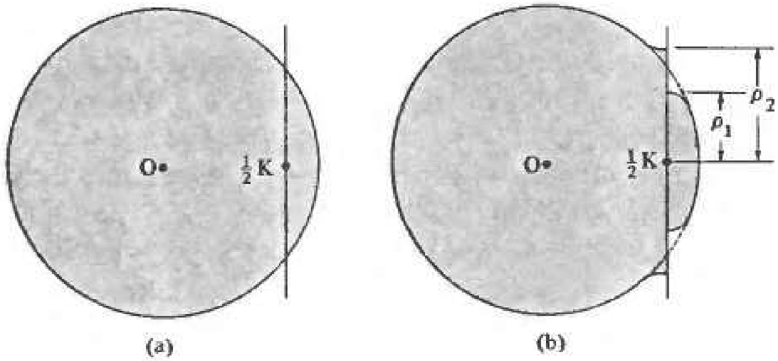

Figure 9.6
(a) Free electron sphere cutting Bragg plane located at $\frac{1}{2}\mathbf{K}$ from the origin ($U_{\mathbf{K}} = 0$). (b) Deformation of the free electron sphere near the Bragg plane when $U_{\mathbf{K}} \neq 0$. The constant-energy surface intersects the plane in two circles, whose radii are calculated in Problem 1.

[11] This follows from the demonstration on page 159 that a constant-energy surface is perpendicular to a Bragg plane when they intersect, in the nearly free electron approximation.

reciprocal lattice point. Since Bragg planes bisect the lines joining the origin to points of the reciprocal lattice, one can equally well define the first zone as the set of points that can be reached from the origin without crossing any Bragg planes.[12]

Higher Brillouin zones are simply other regions bounded by the Bragg planes, defined as follows:

The *first Brillouin zone* is the set of points in k-space that can be reached from the origin without crossing *any* Bragg plane. The *second Brillouin zone* is the set of points that can be reached from the first zone by crossing only one Bragg plane. The *$(n + 1)$th Brillouin zone* is the set of points not in the $(n - 1)$th zone that can be reached from the nth zone by crossing only one Bragg plane.

Alternatively, the *nth Brillouin zone* can be defined as the set of points that can be reached from the origin by crossing $n - 1$ Bragg planes, but no fewer.

These definitions are illustrated in two dimensions in Figure 9.7. The surface of the first three zones for the fcc and bcc lattices are shown in Figure 9.8. Both definitions emphasize the physically important fact that the zones are bounded by Bragg planes. Thus they are regions at whose surfaces the effects of a weak periodic potential are important (i.e., first order), but in whose interior the free electron energy levels are only perturbed in second order.

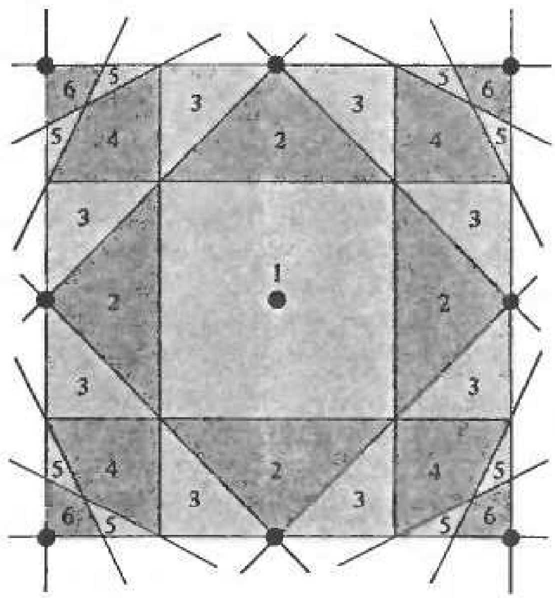

Figure 9.7
Illustration of the definition of the Brillouin zones for a two-dimensional square Bravais lattice. The reciprocal lattice is also a square lattice of side b. The figure shows all Bragg planes (lines, in two dimensions) that lie within the square of side $2b$ centered on the origin. These Bragg planes divide that square into regions belonging to zones 1 to 6. (Only zones 1, 2, and 3 are entirely contained within the square, however.)

It is very important to note that each Brillouin zone is a primitive cell of the reciprocal lattice. This is because the nth Brillouin zone is simply the set of points that have the origin as the nth nearest reciprocal lattice point (a reciprocal lattice point $\mathbf{K}$ is nearer to a point $\mathbf{k}$ than $\mathbf{k}$ is to the origin if and only if $\mathbf{k}$ is separated from the origin by the Bragg plane determined by $\mathbf{K}$). Given this, the proof that the nth Brillouin zone is a primitive cell is identical to the proof on page 73 that the Wigner-Seitz cell (i.e., the first Brillouin zone) is primitive, provided that the phrase "nth nearest neighbor" is substituted for "nearest neighbor" throughout the argument.

[12] We exclude from consideration points lying *on* Bragg planes, which turn out to be points common to the surface of two or more zones. We define the zones in terms of their interior points.

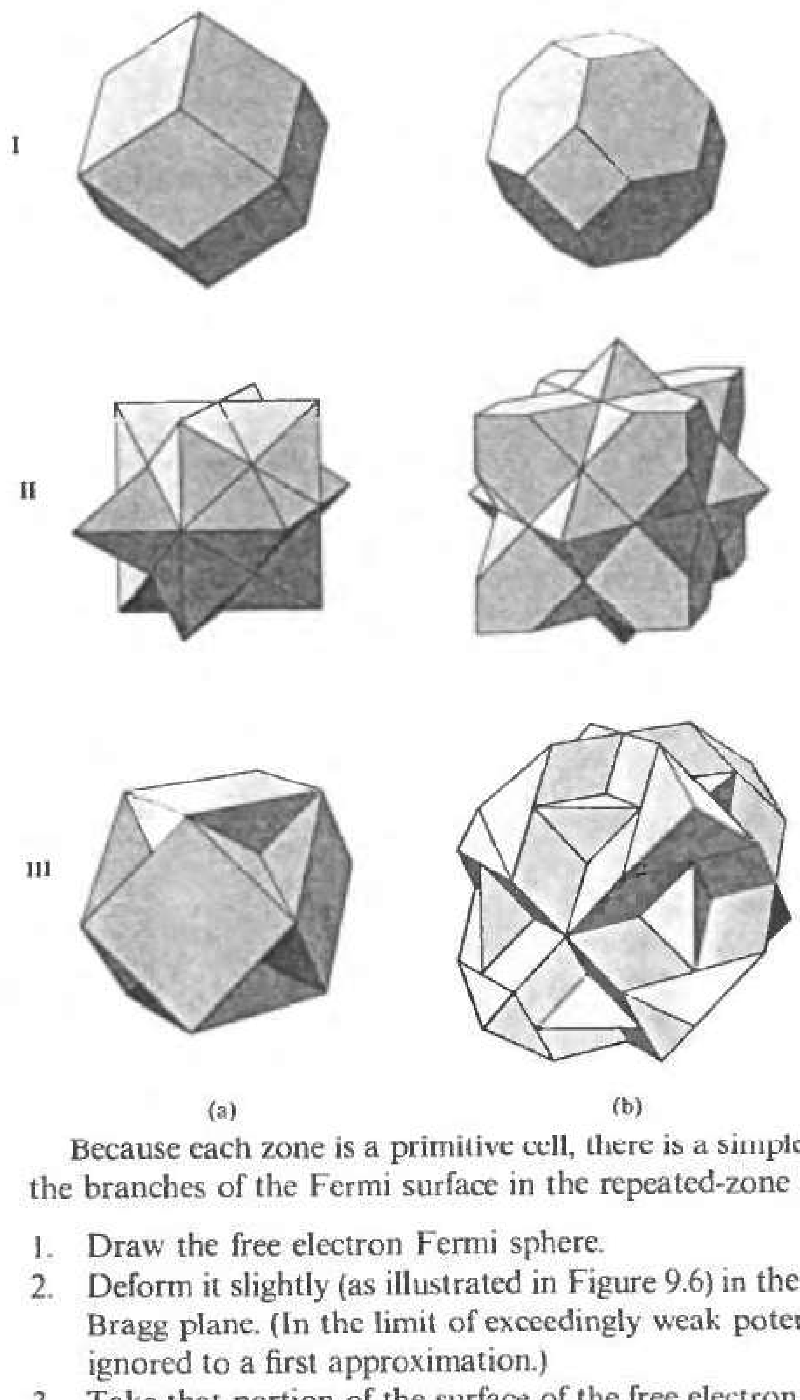

Figure 9.8
Surfaces of the first, second, and third Brillouin zones for (a) body-centered cubic and (b) face-centered cubic crystals. (Only the *exterior* surfaces are shown. It follows from the definition on page 163 that the *interior* surface of the nth zone is identical to the exterior surface of the $(n-1)$th zone.) Evidently the surfaces bounding the zones become increasingly complex as the zone number increases. In practice it is often simplest to construct free electron Fermi surfaces by procedures (such as those described in Problem 4) that avoid making use of the explicit form of the Brillouin zones. (After R. Lück, doctoral dissertation, Technische Hochschule, Stuttgart, 1965.)

Because each zone is a primitive cell, there is a simple algorithm for constructing the branches of the Fermi surface in the repeated-zone scheme[13]:

1. Draw the free electron Fermi sphere.
2. Deform it slightly (as illustrated in Figure 9.6) in the immediate vicinity of every Bragg plane. (In the limit of exceedingly weak potentials this step is sometimes ignored to a first approximation.)
3. Take that portion of the surface of the free electron sphere lying within the nth Brillouin zone, and translate it through all reciprocal lattice vectors. The resulting surface is the branch of the Fermi surface (conventionally assigned to the nth band) in the repeated-zone scheme.[14]

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[13] The representation of the Fermi surface in the repeated-zone scheme is the most general. After surveying each branch in its full periodic splendor, one can pick that primitive cell which most lucidly represents the topological structure of the whole (which is often, but by no means always, the first Brillouin zone).

[14] An alternative procedure is to translate the pieces of the Fermi surface in the nth zone through those reciprocal lattice vectors that take the pieces of the nth zone in which they are contained, into the first zone. (Such translations exist because the nth zone is a primitive cell.) This is illustrated in Figure 9.9. The Fermi surface in the repeated-zone scheme is then constructed by translating the resulting first zone structures through all reciprocal lattice vectors.

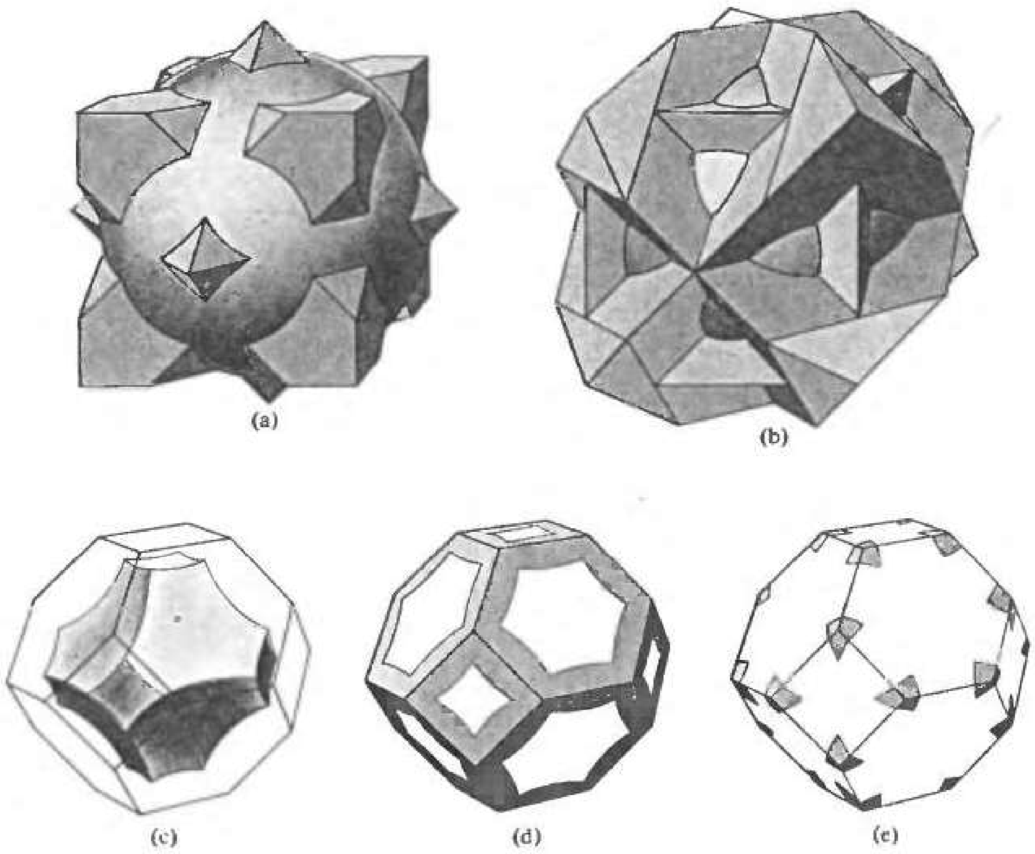

Figure 9.9

The free electron Fermi sphere for a face-centered cubic metal of valence 4. The first zone lies entirely within the interior of the sphere, and the sphere does not extend beyond the fourth zone. Thus the only zone surfaces intersected by the surface of the sphere are the (exterior) surfaces of the second and third zones (cf. Figure 9.8b). The second-zone Fermi surface consists of those parts of the surface of the sphere lying entirely within the polyhedron bounding the second zone (i.e., all of the sphere except the parts extending beyond the polyhedron in (a)). When translated through reciprocal lattice vectors into the first zone, the pieces of the second-zone surface give the simply connected figure shown in (c). (It is known as a "hole surface"; the levels it encloses have higher energies than those outside). The third-zone Fermi surface consists of those parts of the surface of the sphere lying outside of the second zone (i.e., the parts extending beyond the polyhedron in (a)) that do not lie outside the third zone (i.e., that are contained within the polyhedron shown in (b)). When translated through reciprocal lattice vectors into the first zone, these pieces of sphere give the multiply connected structure shown in (d). The fourth-zone Fermi surface consists of the remaining parts of the surface of the sphere that lie outside the third zone (as shown in (b)). When translated through reciprocal lattice vectors into the first zone they form the "pockets of electrons" shown in (e). For clarity (d) and (e) show only the intersection of the third and fourth zone Fermi surfaces with the surface of the first zone. (From R. Lück, *op. cit.*)

Generally speaking, the effect of the weak periodic potential on the surfaces constructed from the free electron Fermi sphere without step 2, is simply to round off the sharp edges and corners. If, however, a branch of the Fermi surface consists of very small pieces of surface (surrounding either occupied or unoccupied levels, known as "pockets of electrons" or "pockets of holes"), then a weak periodic potential may cause these to disappear. In addition, if the free electron Fermi surface has parts with a very narrow cross section, a weak periodic potential may cause it to become disconnected at such points.

Some further constructions appropriate to the discussion of almost free electrons in fcc crystals are illustrated in Figure 9.10. These free-electron-like Fermi surfaces are of great importance in understanding the real Fermi surfaces of many metals. This will be illustrated in Chapter 15.

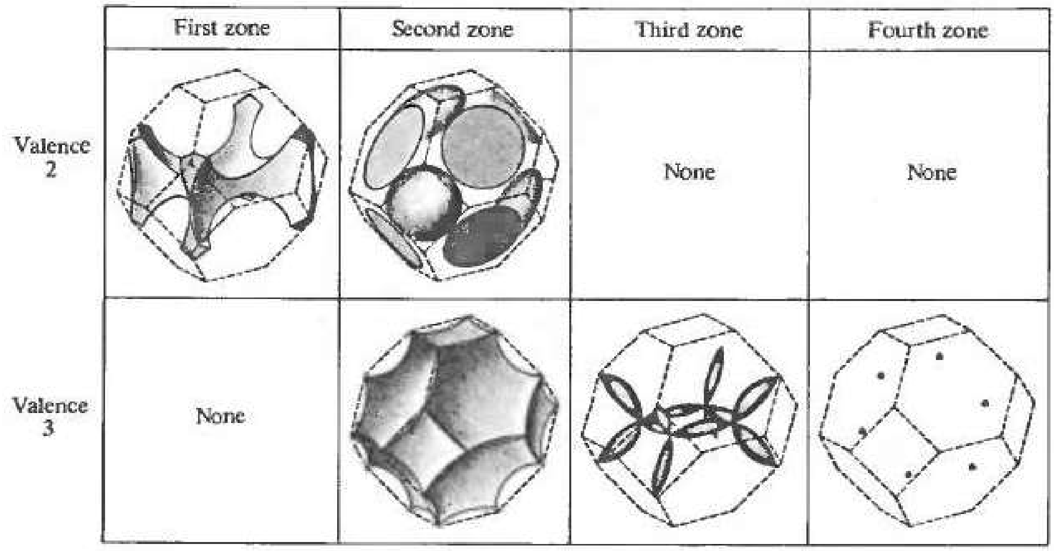

Figure 9.10

The free electron Fermi surfaces for face-centered cubic metals of valence 2 and 3. (For valence 1 the surface lies entirely within the interior of the first zone and therefore remains a sphere to lowest order; the surface for valence 4 is shown in Figure 9.9.) All branches of the Fermi surface are shown. The primitive cells in which they are displayed have the shape and orientation of the first Brillouin zone. However, the cell is actually the first zone (i.e., is centered on $\mathbf{K} = 0$) only in the figures illustrating the second zone surfaces. In the first and third zone figures $\mathbf{K} = 0$ lies at the center of one of the horizontal faces, while for the fourth zone figure it lies at the center of the hexagonal face on the upper right (or the parallel face opposite it (hidden)). The six tiny pockets of electrons constituting the fourth zone surface for valence 3 lie at the corners of the regular hexagon given by displacing that hexagonal face in the [111] direction by half the distance to the face opposite it. (After W. Harrison, *Phys. Rev.* **118**, 1190 (1960).) Corresponding constructions for body-centered cubic metals can be found in the Harrison paper.

THE GEOMETRICAL STRUCTURE FACTOR IN MONATOMIC LATTICES WITH BASES

Nothing said up to now has exploited any properties of the potential $U(\mathbf{r})$ other than its periodicity, and, for convenience, inversion symmetry. If we pay somewhat